Yoochan Lee

Postdoctoral Researcher

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Research Summary

My research is in **systems security**, with a focus on the **Linux kernel**. My goal is to make vulnerabilities **unexploitable in practice**. To this end, I work **offense-to-defense**, uncovering the building blocks that systems unintentionally supply to attackers and developing defenses that eliminate them. This work has appeared at top-tier security venues, including four first-author papers at IEEE S&P, USENIX Security, and NDSS, and reaches beyond academia: I disclose what I find to affected vendors and bring the results to practitioners at venues such as Black Hat.

Academic Appointments

Max Planck Institute for Security and Privacy — Bochum, Germany
Postdoctoral Researcher — *Nov 2025 – Present*
Advisor: Prof. Thorsten Holz

Arizona State University — Tempe, AZ, USA
Visiting Scholar — *Mar 2024 – Jun 2024*

Education

Seoul National University — Seoul, South Korea
M.S./Ph.D. in Electrical and Computer Engineering — *Sep 2019 – Aug 2025*
Advisor: Prof. Byoungyoung Lee
Samsung Electronics MX Research Scholarship, Sep 2023 – Aug 2025

Hanyang University — Seoul, South Korea
B.S. in Computer Science and Engineering — *Mar 2012 – Feb 2018*

Publications

First-Author Refereed Conference Publications

4. **Heap Localization: Cache Side-Channel based Linux Kernel Heap Exploit Techniques**
Yoochan Lee, Sihyun Roh, Hyuk Kwon, Byoungyoung Lee, and Thorsten Holz
*In IEEE Symposium on Security and Privacy (**S&P**), May 2026*

3. **DirtyFree: Simplified Data-Oriented Programming in the Linux Kernel**
Yoochan Lee, Hyuk Kwon, and Thorsten Holz
*In Network and Distributed System Security Symposium (**NDSS**), Feb 2026*

2. **Pspray: Timing Side-Channel based Linux Kernel Heap Exploitation Technique**
Yoochan Lee, Jinhan Kwak, Junesoo Kang, Yuseok Jeon, and Byoungyoung Lee
*In USENIX Security Symposium (**USENIX Security**), Aug 2023*

1. **ExpRace: Exploiting Kernel Races through Raising Interrupts**
Yoochan Lee, Changwoo Min, and Byoungyoung Lee
*In USENIX Security Symposium (**USENIX Security**), Aug 2021*

Other Refereed Conference Publications

3. **GHost in the SHELL: A GPU-to-Host Memory Attack and Its Mitigation**
   Sihyun Roh, Woohyuk Choi, Jaeyoung Chung, **Yoochan Lee**, Suhwan Song, and Byoungyoung Lee
   *In IEEE Symposium on Security and Privacy (**S&P**), May 2026*

2. **PeTAL: Ensuring Access Control Integrity against Data-only Attacks on Linux**
   Juhee Kim, Jinbum Park, **Yoochan Lee**, Chengyu Song, Taesoo Kim, and Byoungyoung Lee
   *In ACM Conference on Computer and Communications Security (**CCS**), Oct 2024*

1. **Diagnosing Kernel Concurrency Failures with AITIA**
   Dae R. Jeong, Minkyu Jung, **Yoochan Lee**, Byoungyoung Lee, Insik Shin, and Youngjin Kwon
   *In European Conference on Computer Systems (**EuroSys**), May 2023*

Submitted Manuscripts

- **Know Your Heap Before You Leap: Blind Reconnaissance Toward Environment-Agnostic Kernel Exploitation**
  Yoochan Lee and Thorsten Holz
  Under submission

- **SlabOrchestra: Environment-Adaptive Heap Manipulation Technique in the Linux Kernel**
  Yoochan Lee, Yonghwa Lee, and Thorsten Holz
  Under submission

Talks

Invited Academic Talks

- **Your CPU Knows Where Your Heap Objects Are**
  Ruhr University Bochum, Germany, Oct 2026

- **Revisiting Well-Known Building Blocks for Practical Linux Kernel Exploitation**
  École Polytechnique Fédérale de Lausanne (EPFL), Switzerland, Apr 2026

Selected Industry and Practitioner Talks

- **Privilege Escalation using DOP in x86-64 macOS**
  Yoochan Lee, Sangjun Song, Junoh Lee, and Jeongsu Choi
  Hack In The Box Amsterdam, Apr 2023

- **Perfect Spray: A Journey From Finding a New Type of Logical Flaw at Linux Kernel To Developing a New Heap Exploitation Technique**
  Yoochan Lee, Jinhan Kwak, Junesoo Kang, Yuseok Jeon, and Byoungyoung Lee
  Black Hat Europe, Dec 2022

- **Exploiting Kernel Races through Taming Thread Interleaving**
  Yoochan Lee, Changwoo Min, and Byoungyoung Lee
  Black Hat USA, Aug 2020

Professional Service

Technical Program Committee, USENIX Security Symposium 2027

External Reviewer: USENIX Security 2021, 2022, 2025; IEEE S&P 2024, 2025; ACM CCS 2025; NDSS 2020; ACSAC 2023

Teaching and Mentoring

Teaching Assistant

Cyber Security and Blockchain (M2177.006300) Spring 2020, Fall 2020, Fall 2021, Fall 2022
Seoul National University

Programming Methodology (430.211) Spring 2025
Seoul National University

Mentoring

Best of the Best (BoB) — Mentor Jul 2023 – Feb 2026
Selective national cybersecurity talent program run by KITRI under the Ministry of Science and ICT, South Korea

- Mentored three student project teams comprising 18 students through the program's multi-month project phase, advising on kernel exploitation and guiding responsible disclosure of real-world vulnerabilities (e.g., Linux, NVIDIA).

White Hat School — Lead Mentor Sep 2023 – Dec 2025
Government-sponsored cybersecurity training program for early-career talent, run by KITRI, South Korea

- Designed the **program curriculum** and delivered its **systems security** lectures and hands-on labs to several hundred students per cohort, then mentored student teams through their multi-month projects.

Selected Vulnerability Discoveries and Responsible Disclosure

Solidly smart contract (2022): Critical vulnerability enabling unauthorized fund withdrawal.
CVE-2021-31077 (macOS): Kernel heap overflow leading to local privilege escalation.
CVE-2018-4417 (macOS): Kernel information leakage.
CVE-2018-4338 (macOS): Kernel information leakage.
CVE-2018-4084 (macOS): Kernel information leakage.
CVE-2017-7014 (macOS): Arbitrary kernel code execution.

Honors and Awards (Selected)

- **Merit Award**, White Hacker's Night, Dec 2023 (contributions to cybersecurity education)
- **Multiple-time DEF CON CTF Finalist** (2017–2022), Las Vegas, USA
  3rd Place, DEF CON 30 CTF (2022); **4th Place**, DEF CON 29 CTF (2021)
- **1st Place**, Cyber Conflict Exercise and Contest 2018 (Team GYG), Jeju, South Korea, Oct 2018
- **1st Place**, Secuinside Capture The Bug (Team Minionz), Seoul, South Korea, Jul 2016
- **Top 10**, Best of the Best (BoB), 4th Cohort, Mar 2016

Industry Experience

Raon WhiteHat, Seoul, South Korea Feb–Aug 2017
Security Intern — Penetration Testing

Naver Labs, Gyeonggi-do, South Korea Apr–Jun 2016
Security Intern — Browser Vulnerability Research (Naver Whale)

Electronics and Telecommunications Research Institute, Daejeon, Korea Dec 2014–Feb 2015
Intern — Network Security Team

References

Byoungyoung Lee	Seoul National University	byoungyoung@snu.ac.kr
Thorsten Holz	Max Planck Institute for Security and Privacy	thorsten.holz@mpi-sp.org
Yuseok Jeon	Korea University	ys_jeon@korea.ac.kr